**Title: 2D House Model**

```
#include <GL/glut.h>
#include <GL/gl.h>
#include <GL/glu.h>
#include <math.h>

/* -------- Shape Functions -------- */
void drawRect(float x1, float y1, float x2, float y2, int filled)
{
   if (filled)
      glBegin(GL_QUADS);
   else
      glBegin(GL_LINE_LOOP);

   glVertex2f(x1, y1);
   glVertex2f(x2, y1);
   glVertex2f(x2, y2);
   glVertex2f(x1, y2);
   glEnd();
}
void drawTriangle(float x1, float y1, float x2, float y2, float x3, float y3, int filled)
{
   if (filled)
      glBegin(GL_TRIANGLES);
   else
      glBegin(GL_LINE_LOOP);

   glVertex2f(x1, y1);
   glVertex2f(x2, y2);
   glVertex2f(x3, y3);
   glEnd();
}
void drawCircle(float cx, float cy, float r, int segments, int filled)
{
   float theta, x, y;
   if (filled) glBegin(GL_TRIANGLE_FAN);
   else
      glBegin(GL_LINE_LOOP);

   if (filled)
      glVertex2f(cx, cy);

   for (int i = 0; i <= segments; i++)
   {
      theta = 2.0f * 3.1415926f * i / segments;
      x = r * cosf(theta);
      y = r * sinf(theta);
      glVertex2f(cx + x, cy + y);
   }
   glEnd();
}

/* -------- Initialization -------- */
void init(void)
{
   glClearColor(0.75f, 0.90f, 1.0f, 1.0f);
   glMatrixMode(GL_PROJECTION);
   glLoadIdentity();
   gluOrtho2D(0.0, 100.0, 0.0, 100.0);
}
/* -------- Display Function -------- */
void display(void)
{
   glClear(GL_COLOR_BUFFER_BIT);

   /* Ground */
   glColor3f(0.2f, 0.7f, 0.2f);
   drawRect(0, 0, 100, 25, 1);

   /* House Walls */
   glColor3f(0.95f, 0.75f, 0.55f);
   drawRect(25, 25, 75, 65, 1);

   /* Wall outline */
   glColor3f(0.2f, 0.2f, 0.2f);
   drawRect(25, 25, 75, 65, 0);

   /* Roof */
   glColor3f(0.8f, 0.2f, 0.2f);
   drawTriangle(20, 65, 50, 88, 80, 65, 1);

   /* Roof outline */
   glColor3f(0.2f, 0.2f, 0.2f);
   drawTriangle(20, 65, 50, 88, 80, 65, 0);

   /* Door */
   glColor3f(0.35f, 0.2f, 0.1f);
   drawRect(46, 25, 54, 48, 1);

   glColor3f(0.1f, 0.1f, 0.1f);
   drawRect(46, 25, 54, 48, 0);

   /* Door knob */
   glColor3f(1.0f, 0.85f, 0.0f);
   drawCircle(52.5f, 36.0f, 1.0f, 40, 1);

   /* Left Window */
   glColor3f(0.7f, 0.9f, 1.0f);
   drawRect(30, 40, 40, 52, 1);

   glColor3f(0.1f, 0.1f, 0.1f);
   drawRect(30, 40, 40, 52, 0);

   glBegin(GL_LINES);
```

```
   glVertex2f(35, 40); glVertex2f(35, 52);
   glVertex2f(30, 46); glVertex2f(40, 46);
   glEnd();

   /* Right Window */
   glColor3f(0.7f, 0.9f, 1.0f);
   drawRect(60, 40, 70, 52, 1);

   glColor3f(0.1f, 0.1f, 0.1f);
   drawRect(60, 40, 70, 52, 0);

   glBegin(GL_LINES);
   glVertex2f(65, 40); glVertex2f(65, 52);
   glVertex2f(60, 46); glVertex2f(70, 46);
   glEnd();

   glutSwapBuffers();
}

/* -------- Main Function -------- */
int main(int argc, char **argv)
{
   glutInit(&argc, argv);
   glutInitDisplayMode(GLUT_RGB | GLUT_DOUBLE);

   glutInitWindowPosition(100, 100);
   glutInitWindowSize(700, 700);

   glutCreateWindow("2D House Design (C + OpenGL)");

   init();
   glutDisplayFunc(display);
   glutMainLoop();
   return 0;
}
```

**Title: Moving Object**

```
#include <windows.h>
#include <GL/glut.h>
#include <stdlib.h>
#include <math.h>
float p = -10.0f;

/* -------- Display Function -------- */
void display(void)
{
   glClear(GL_COLOR_BUFFER_BIT);

   /* Update position */
   if (p <= 10)
      p += 0.005f;
   else
      p = -10.0f;

   /* Draw moving rectangle */
   glBegin(GL_QUADS);
      glColor3f(1.0f, 1.0f, 1.0f);
      glVertex2f(p, 3);
      glVertex2f(p + 3, 3);
      glVertex2f(p + 3, -3);
      glVertex2f(p, -3);
   glEnd();
   glFlush();

   /* Request next frame (animation loop) */
   glutPostRedisplay();
}

/* -------- Initialization -------- */
void init(void)
{
   glClearColor(1.0f, 0.0f, 0.0f, 0.0f);

   glMatrixMode(GL_PROJECTION);
   glLoadIdentity();
   glOrtho(-10, 10, -10, 10, -10, 10);
}

/* -------- Main Function -------- */
int main(int argc, char** argv)
{
   glutInit(&argc, argv);
   glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);

   glutInitWindowSize(600, 600);
   glutInitWindowPosition(100, 100);
   glutCreateWindow("Moving Object");
   init();
   glutDisplayFunc(display);
   glutMainLoop();
   return 0;
}
```

**Title: Circle using Mid-point**

```
#include <GL/glut.h>
#include <stdio.h>
#include <math.h>

/* Circle center and radius */
int xc = 0, yc = 0, r = 20;
/* -------- Plot Symmetric Points -------- */
void plotPoints(int x, int y)
{
```

```
    glVertex2i(xc + x, yc + y);
    glVertex2i(xc - x, yc + y);
    glVertex2i(xc + x, yc - y);
    glVertex2i(xc - x, yc - y);
    glVertex2i(xc + y, yc + x);
    glVertex2i(xc - y, yc + x);
    glVertex2i(xc + y, yc - x);
    glVertex2i(xc - y, yc - x);
}

/* -------- Display Function -------- */
void display()
{
    glClear(GL_COLOR_BUFFER_BIT);
    glColor3f(1.0f, 0.0f, 0.0f);
    glPointSize(3.0);   // Make points visible
    glBegin(GL_POINTS);

    int x = 0;
    int y = r;
    int p = 1 - r;
    plotPoints(x, y);
    while (x < y)
    {
        x++;
        if (p < 0) {
            p = p + 2 * x + 1;
        }
        else {
            y--;
            p = p + 2 * x - 2 * y + 1;
        }
        plotPoints(x, y);
    }
    glEnd();
    glFlush();
}

/* -------- Initialization -------- */
void init()
{
    glClearColor(0.0f, 0.0f, 0.0f, 0.0f);
    glMatrixMode(GL_PROJECTION);
    glLoadIdentity();
    glOrtho(-50.0, 50.0, -50.0, 50.0, -1.0, 1.0);
}

/* -------- Main Function -------- */
int main(int argc, char** argv)
{
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);
    glutInitWindowSize(500, 500);
    glutInitWindowPosition(100, 100);
    glutCreateWindow("Midpoint Circle Algorithm");
    init();
    glutDisplayFunc(display);
    glutMainLoop();
    return 0;
}
```

**Title: Shaheed Minar, a bicycle, and the national flag**

```
#include <GL/glut.h>
#include <math.h>

/* -------- Utility Shapes -------- */
void drawCircle(float cx, float cy, float r)
{
    glBegin(GL_POLYGON);
    for (int i = 0; i < 100; i++)
    {
        float theta = 2.0f * 3.1416f * i / 100;
        float x = r * cosf(theta);
        float y = r * sinf(theta);
        glVertex2f(cx + x, cy + y);
    }
    glEnd();
}

void drawRect(float x1, float y1, float x2, float y2)
{
    glBegin(GL_QUADS);
    glVertex2f(x1, y1);
    glVertex2f(x2, y1);
    glVertex2f(x2, y2);
    glVertex2f(x1, y2);
    glEnd();
}

void drawLine(float x1, float y1, float x2, float y2)
{
    glBegin(GL_LINES);
    glVertex2f(x1, y1);
    glVertex2f(x2, y2);
    glEnd();
}

/* -------- Bangladesh Flag -------- */
void drawFlag()
{
    glColor3f(0.4f, 0.2f, 0.1f); // pole
    drawRect(60, 150, 65, 350);

    glColor3f(0.0f, 0.6f, 0.0f); // green
    drawRect(65, 280, 220, 350);
```

```
   glColor3f(1.0f, 0.0f, 0.0f); // red
circle
   drawCircle(140, 315, 30);
}

/* -------- Shaheed Minar -------- */
void drawShaheedMinar()
{
   glColor3f(0.8f, 0.8f, 0.8f);

   // base
   drawRect(300, 100, 500, 120);
   drawRect(320, 120, 480, 140);

   // pillars
   drawRect(330, 140, 345, 280);
   drawRect(360, 140, 375, 310);
   drawRect(395, 140, 415, 340);
   drawRect(430, 140, 445, 310);
   drawRect(460, 140, 475, 280);

   // top bar
   drawRect(330, 280, 475, 290);

   // red circle
   glColor3f(1.0f, 0.0f, 0.0f);
   drawCircle(400, 220, 35);
}

/* -------- Bicycle -------- */
void drawBicycle()
{
   glColor3f(0.0f, 0.0f, 0.0f);

   // wheels
   drawCircle(620, 120, 40);
   drawCircle(760, 120, 40);

   glLineWidth(3);
   // frame
   drawLine(620, 120, 680, 180);
   drawLine(680, 180, 720, 120);
   drawLine(620, 120, 720, 120);
   drawLine(680, 180, 760, 120);

   // seat
   drawLine(680, 180, 700, 220);
   drawLine(700, 220, 740, 220);

   // handle
   drawLine(700, 220, 730, 260);
   drawLine(730, 260, 760, 250);
}

/* -------- Display -------- */
void display()
{
   glClear(GL_COLOR_BUFFER_BIT);

   drawFlag();
   drawShaheedMinar();
   drawBicycle();

   glFlush();
}

/* -------- Init -------- */
void init()
{
   glClearColor(1.0f, 1.0f, 1.0f, 1.0f);

   glMatrixMode(GL_PROJECTION);
   glLoadIdentity();
   gluOrtho2D(0, 900, 0, 500);
}

/* -------- Main -------- */
int main(int argc, char **argv)
{
   glutInit(&argc, argv);
   glutInitDisplayMode(GLUT_SINGLE
| GLUT_RGB);
   glutInitWindowSize(900, 500);
   glutInitWindowPosition(100, 100);
   glutCreateWindow("Shaheed
Minar, Bicycle, and Bangladesh
Flag");
   init();
   glutDisplayFunc(display);
   glutMainLoop();
   return 0;
}
```

**Title: Circle,half-circle,polygon**

```
#include <GL/glut.h>
#include <math.h>
#include <stdio.h>

/* -------- Initialization -------- */
void init(void)
{
   glClearColor(0.0f, 0.0f, 0.0f, 1.0f);
   glMatrixMode(GL_PROJECTION);
   glLoadIdentity();
   glOrtho(0.0, 50.0, 0.0, 50.0, -1.0,
1.0);
}

/* -------- Draw Full Circle -------- */
void drawCircle(float cx, float cy,
float r)
{
   glBegin(GL_LINE_LOOP);
```

```
    for (int i = 0; i < 100; i++)
    {
        float theta = 2.0f * 3.1416f * i / 100.0f;
        float x = r * cosf(theta);
        float y = r * sinf(theta);
        glVertex2f(cx + x, cy + y);
    }
    glEnd();
}

/* -------- Draw Half Circle -------- */
void drawHalfCircle(float cx, float cy, float r)
{
    glBegin(GL_LINE_STRIP);
    for (int i = 0; i <= 50; i++) {
        float theta = 3.1416f * i / 50.0f;
        float x = r * cosf(theta);
        float y = r * sinf(theta);
        glVertex2f(cx + x, cy + y);
    }
    glEnd();
}

/* -------- Draw Polygon -------- */
void drawPolygon()
{
    glBegin(GL_POLYGON);

    glVertex2f(34, 10);
    glVertex2f(42, 15);
    glVertex2f(39, 24);
    glVertex2f(29, 24);
    glVertex2f(26, 15);
    glEnd();
}

/* -------- Display Function -------- */
void display()
{
    glClear(GL_COLOR_BUFFER_BIT);

    /* Circle */
    glColor3f(1.0f, 0.0f, 0.0f);
    drawCircle(10, 25, 6);

    /* Half Circle */
    glColor3f(0.0f, 1.0f, 0.0f);
    drawHalfCircle(25, 25, 6);

    /* Polygon */
    glColor3f(0.0f, 0.0f, 1.0f);
    drawPolygon();
    glutSwapBuffers();
}

/* -------- Main Function -------- */
int main(int argc, char **argv)
{
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_RGB | GLUT_DOUBLE);
    glutInitWindowSize(500, 500);
    glutInitWindowPosition(100, 100);
    glutCreateWindow("Drawing Basic Shapes");
    init();
    glutDisplayFunc(display);
    glutMainLoop();
    return 0;
}
```

**Title: 4 stars**

```
#include <GL/glut.h>
#include <stdio.h>
#include <GL/gl.h>
#include <math.h>

void init(void)
{
glClearColor(0.0, 0.0, 0.0, 0.0);
glMatrixMode(GL_PROJECTION);
glLoadIdentity();
glOrtho(0.0, 20.0, 0.0, 20.0, -1.0, 1.0);
}
void Draw()
{
glClear(GL_COLOR_BUFFER_BIT);
// RED STAR
glColor3f(1.0f, 0.0f, 0.0f);
glBegin(GL_POLYGON);
glVertex2f(10,10);
glVertex2f(7,11);
glVertex2f(7,9);
glVertex2f(7,9);
glVertex2f(5,9);
glVertex2f(5,11);
glVertex2f(7,11);
glEnd();
glBegin(GL_POLYGON);
glVertex2f(5,11);
glVertex2f(6,14);
glVertex2f(7,11);
glEnd();
glBegin(GL_POLYGON);
glVertex2f(5,11);
glVertex2f(2,10);
glVertex2f(5,9);
```

```
glEnd();
glBegin(GL_POLYGON);

glVertex2f(5,9);
glVertex2f(6,6);
glVertex2f(7,9);
glEnd();
// WHITE STAR
glColor3f(1.0f, 1.0f, 1.0f);
glBegin(GL_POLYGON);
glVertex2f(10,10);
glVertex2f(13,11);
glVertex2f(13,9);
glVertex2f(13,9);
glVertex2f(15,9);
glVertex2f(15,11);
glVertex2f(13,11);
glEnd();
glBegin(GL_POLYGON);
glVertex2f(13,9);
glVertex2f(14,6);
glVertex2f(15,9);
glEnd();
glBegin(GL_POLYGON);
glVertex2f(15,11);
glVertex2f(18,10);
glVertex2f(15,9);
glEnd();
glBegin(GL_POLYGON);
glVertex2f(13,11);
glVertex2f(14,14);
glVertex2f(15,11);
glEnd();
// BLUE STAR
glColor3f(0.0f, 0.0f, 1.0f);

glBegin(GL_POLYGON);
glVertex2f(10,10);
glVertex2f(9,7);
glVertex2f(11,7);
glVertex2f(9,7);
glVertex2f(9,5);
glVertex2f(11,5);
glVertex2f(11,7);
glEnd();
glBegin(GL_POLYGON);
glVertex2f(9,7);
glVertex2f(6,6);
glVertex2f(9,5);
glEnd();
glBegin(GL_POLYGON);
glVertex2f(9,5);
glVertex2f(10,2);
glVertex2f(11,5);
glEnd();
glBegin(GL_POLYGON);
glVertex2f(11,5);
glVertex2f(14,6);
glVertex2f(11,7);
glEnd();
// GREEN STAR
glColor3f(0.0f, 1.0f, 0.0f);
glBegin(GL_POLYGON);
glVertex2f(10,10);
glVertex2f(9,13);
glVertex2f(11,13);
glVertex2f(9,13);
glVertex2f(9,15);
glVertex2f(11,15);

glVertex2f(11,13);
glEnd();
glBegin(GL_POLYGON);
glVertex2f(9,13);
glVertex2f(6,14);
glVertex2f(9,15);
glEnd();
glBegin(GL_POLYGON);
glVertex2f(9,15);
glVertex2f(10,18);
glVertex2f(11,15);
glEnd();
glBegin(GL_POLYGON);
glVertex2f(11,15);
glVertex2f(14,14);
glVertex2f(11,13);
glEnd();
glFlush();
glutSwapBuffers();
}
int main(int argc, char **argv)
{
glutInit(&argc, argv);
glutInitDisplayMode(GLUT_RGB |
GLUT_DOUBLE);
glutInitWindowPosition(0, 0);
glutInitWindowSize(600, 600);
glutCreateWindow("4 STAR");
init();
glutDisplayFunc(Draw);
glutMainLoop();
return 0;
}
```

**Title: You have to draw objects in one scene using basic shapes like lines, quadrilaterals, triangles, circles, and polygons.**

```
#include <GL/glut.h>
#include <math.h>

/* -------- Utility Shapes -------- */
void drawCircle(float cx, float cy,
float r)
```

```
{
    glBegin(GL_POLYGON);
    for (int i = 0; i < 100; i++)
    {
        float theta = 2.0f * 3.1416f * i /
100;
        float x = r * cosf(theta);
        float y = r * sinf(theta);
        glVertex2f(cx + x, cy + y);
    }
    glEnd();
}

void drawRect(float x1, float y1, float
x2, float y2)
{
    glBegin(GL_QUADS);
    glVertex2f(x1, y1);
    glVertex2f(x2, y1);
    glVertex2f(x2, y2);
    glVertex2f(x1, y2);
    glEnd();
}

void drawTriangle(float x1,float
y1,float x2,float y2,float x3,float y3)
{
    glBegin(GL_TRIANGLES);
    glVertex2f(x1,y1);
    glVertex2f(x2,y2);
    glVertex2f(x3,y3);
    glEnd();
}

void drawLine(float x1,float y1,float
x2,float y2)
{
    glBegin(GL_LINES);
    glVertex2f(x1,y1);
    glVertex2f(x2,y2);
    glEnd();
}

/* -------- Scene Objects -------- */
void drawHouse()
{
    // wall
    glColor3f(0.9,0.7,0.5);
    drawRect(30,20,60,45);

    // roof
    glColor3f(0.8,0.2,0.2);
    drawTriangle(25,45,45,65,65,45);

    // door
    glColor3f(0.3,0.2,0.1);
    drawRect(42,20,48,35);
}

void drawTree()
{
    // trunk
    glColor3f(0.4,0.2,0.1);
    drawRect(70,20,75,40);

    // leaves
    glColor3f(0.0,0.6,0.0);
    drawCircle(72.5,50,10);
}

void drawSun()
{
    glColor3f(1.0,1.0,0.0);
    drawCircle(80,80,8);
    // rays (lines)
    for(int i=0;i<360;i+=30)
    {
        float x = 80 +
12*cos(i*3.1416/180);
        float y = 80 +
12*sin(i*3.1416/180);
        drawLine(80,80,x,y);
    }
}

void drawBicycle()
{
    glColor3f(0,0,0);
    // wheels
    drawCircle(20,20,6);
    drawCircle(35,20,6);

    // frame
    drawLine(20,20,27,30);
    drawLine(27,30,35,20);
    drawLine(20,20,35,20);
}

/* -------- Display -------- */
void display()
{
    glClear(GL_COLOR_BUFFER_BIT);

    // sky
    glColor3f(0.5,0.8,1.0);
    drawRect(0,50,100,100);

    // ground
    glColor3f(0.2,0.7,0.2);
    drawRect(0,0,100,50);
```

```
   drawSun();
   drawHouse();
   drawTree();
   drawBicycle();

   glFlush();
}

/* -------- Init -------- */
void init()
{
   glClearColor(1,1,1,1);

   glMatrixMode(GL_PROJECTION);
   glLoadIdentity();
   gluOrtho2D(0,100,0,100);
}

/* -------- Main -------- */
int main(int argc, char** argv)
{
   glutInit(&argc,argv);
   glutInitDisplayMode(GLUT_SINGLE
| GLUT_RGB);

   glutInitWindowSize(700,700);
   glutCreateWindow("Scene
Drawing using Basic Shapes");

   init();
   glutDisplayFunc(display);

   glutMainLoop();
   return 0;
}
```

**Title: Basic shape**

```
#include <GL/glut.h>
#include <math.h>

/* -------- Circle -------- */
void drawCircle(float cx, float cy,
float r)
{
   glBegin(GL_POLYGON);
   for (int i = 0; i < 100; i++)
   {
      float theta = 2.0f * 3.1416f * i /
100;
      float x = r * cosf(theta);
      float y = r * sinf(theta);
      glVertex2f(cx + x, cy + y);
   }
   glEnd();
}

/* -------- Display -------- */
void display()
{
   glClear(GL_COLOR_BUFFER_BIT);

   /* -------- Line -------- */
   glColor3f(1.0f, 0.0f, 0.0f);
   glBegin(GL_LINES);
   glVertex2f(5, 45);
   glVertex2f(20, 45);
   glEnd();

   /* -------- Quadrilateral -------- */
   glColor3f(0.0f, 1.0f, 0.0f);
   glBegin(GL_QUADS);
   glVertex2f(5, 25);
   glVertex2f(20, 25);
   glVertex2f(20, 35);
   glVertex2f(5, 35);
   glEnd();

   /* -------- Triangle -------- */
   glColor3f(0.0f, 0.0f, 1.0f);
   glBegin(GL_TRIANGLES);
   glVertex2f(30, 25);
   glVertex2f(40, 40);
   glVertex2f(50, 25);
   glEnd();

   /* -------- Circle -------- */
   glColor3f(1.0f, 1.0f, 0.0f);
   drawCircle(70, 35, 8);

   /* -------- Polygon (Pentagon) -------
- */
   glColor3f(1.0f, 0.0f, 1.0f);
   glBegin(GL_POLYGON);
   glVertex2f(60, 10);
   glVertex2f(70, 15);
   glVertex2f(65, 25);
   glVertex2f(55, 25);
   glVertex2f(50, 15);
   glEnd();

   glFlush();
}

/* -------- Init -------- */
void init()
{
   glClearColor(0.0f, 0.0f, 0.0f, 1.0f);

   glMatrixMode(GL_PROJECTION);
   glLoadIdentity();
   gluOrtho2D(0, 100, 0, 50);
```

```
}

/* -------- Main -------- */
int main(int argc, char** argv)
{
   glutInit(&argc, argv);
   glutInitDisplayMode(GLUT_SINGLE
| GLUT_RGB);

   glutInitWindowSize(700, 400);
   glutCreateWindow("Basic Shapes
in One Scene");

   init();
   glutDisplayFunc(display);

   glutMainLoop();
   return 0;
}
```